

Task 4: Weather pattern Analysis

Objective

To design and develop an interactive dashboard for visualizing data, monitoring key performance indicators (KPIs), and generating analytical insights for decision-making.

Tools Used

Python

Streamlit

Pandas

Matplotlib

Numpy

Features

1. Interactive Dashboard Interface
2. KPI Monitoring
3. Statistical Summary
4. Comparative Analysis

Methodology

The Collect and organize data.Process data using Pandas.Calculate KPIs and statistical measures.

Create charts and graphs using Matplotlib.Display results through a Streamlit dashboard.

Expected Output

Interactive charts, data tables, and summary metrics that help users understand trends and performance.

Conclusion

The Advanced Data Analytics Dashboard provides meaningful insights through interactive visualizations and statistical analysis. It helps users monitor performance, identify trends, and make informed decisions efficiently.